IFC Cambricon-LLM Inference Performance

Absolute decode throughput and token latency from checked simulator artifacts.

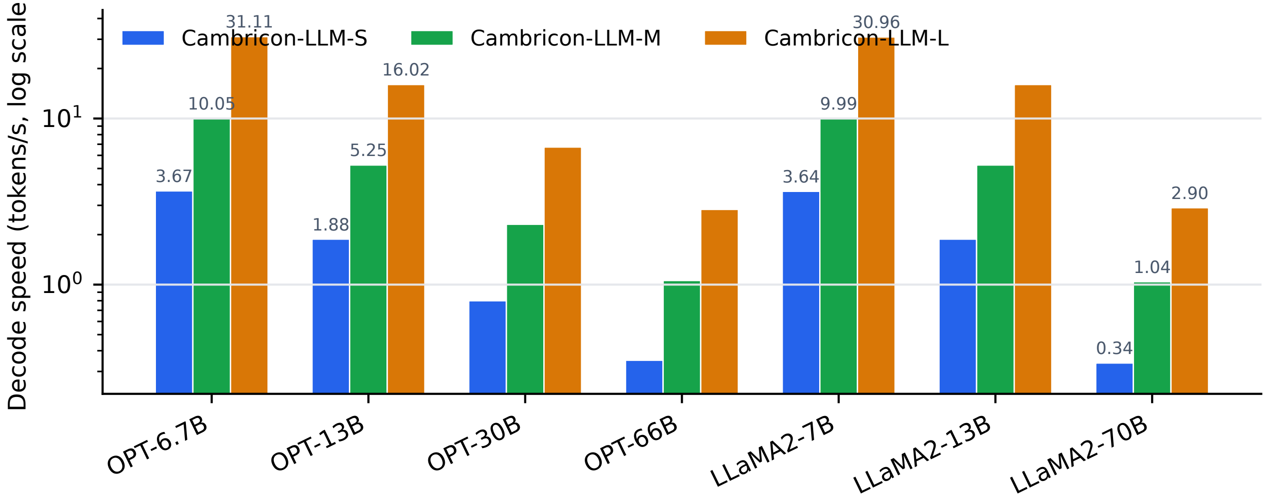
Fastest simulated decode
31.115 token/s
OPT-6.7B on Cambricon-LLM-L

LLaMA2-7B / L
30.959 token/s
32.301 ms/token

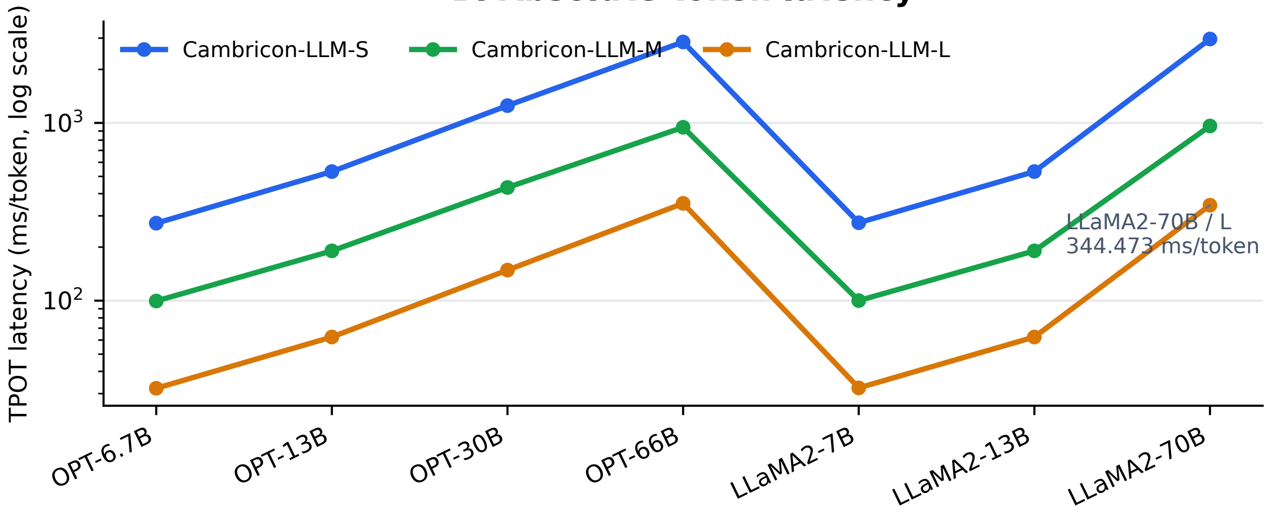
LLaMA2-70B / L
2.903 token/s
344.473 ms/token

Figure 9 W8A8 coverage
21 points
OPT + LLaMA2, S/M/L scales

A. Absolute simulated decode throughput



B. Absolute token latency



C. Absolute throughput table (tokens/s)

	Cambricon-LLM-S	Cambricon-LLM-M	Cambricon-LLM-L
OPT-6.7B	3.666	10.047	31.115
OPT-13B	1.878	5.247	16.017
OPT-30B	0.799	2.308	6.731
OPT-66B	0.350	1.059	2.836
LLaMA2-7B	3.644	9.991	30.959
LLaMA2-13B	1.878	5.247	16.017
LLaMA2-70B	0.337	1.041	2.903

D. Modeling checks used for the released results

Check	Result
Read slicing, Cambricon-LLM-S	**1.683x-1.699x simulated speedup** Paper reference window: 1.6x-1.8x
Hardware-aware tiling, Cambricon-LLM-S	**1.341x-1.349x simulated speedup** Paper reference window: 1.3x-1.4x
SystemC replay checker	**0-cycle final-time delta** 1,536 events match the C event backend
SystemC component model	**+85.5 ns final-time delta** 0.027039% vs the C command-cycle backend

Reference-fit metrics remain in results/summary.json; this dashboard emphasizes absolute simulator performance.

Source: checked CSV/JSON artifacts under results/. Throughput is decode token/s; TPOT is time per output token.